Chapter 4

Modeling Causal Impact via Randomized Experiments

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Just as the rapid adoption of the telephone sparked Erlang's stochastic models for network congestion at the turn of the 20th century, the severe food shortages in 1920s Europe following World War I pushed statisticians to develop better methods for evaluating crop varieties and soil conditions to improve agricultural yields.

Enter the protagonist of this chapter, Jerzy Neyman, who from 1917 to 1921 served as a statistician at the Institute of Agriculture in Bydgoszcz, Poland. His seminal work in 1923 would lay the foundation for modern causal inference, the scientific and business impact of which can hardly be understated.

From the viewpoint of modeling, Neyman's overarching goal was remarkably similar to Erlang's: to develop an effective model of a complex reality that, when calibrated with observational data, would provide causal guidance on how to achieve better outcomes. However, a closer inspection reveals some striking and fascinating divergences between the two approaches:

1. Whereas Erlangian models are highly mechanistic, we will see that Neyman's causal inference framework does not assume we can (or even need to) accurately describe those underlying mechanisms. Rather than modeling reality as interacting submodules governed by explicit rules, Neyman takes a more end-to-end approach, directly estimating the causal impact of an intervention instead of laboring to map the inner workings of the system in question.

2. Whereas Erlangian models tend to rely on a deterministic core structure (such as the mechanisms governing buffers and servers) to describe a highly stochastic system, the original Neymanian framing is almost the opposite: it aims to estimate *deterministic* treatment effects by intentionally introducing randomness into the assignment process.

We will begin this chapter by walking through a basic version of Neyman's result using more modern notation and terminology developed by Rubin. We will then take a deeper dive into these two divergences and see what we can learn from this new way of thinking about impact and causality.

4.1 Learning Treatment Effects via Randomization

Potential outcomes.

Consider an RCT with n units, indexed by $i = 1, \ldots, n$. Each unit receives a binary treatment $W_i \in \{0, 1\}$ and yields an observed outcome Y_i. For each unit, define potential outcomes $Y_i(1)$ and $Y_i(0)$, linked to the observed outcome by

$$Y_i = Y_i(W_i).$$

For unit i, we can define the individual treatment effect as

$$\Delta_i = Y_i(1) - Y_i(0),$$

that is, the difference in outcomes between choosing the treatment and choosing the control intervention. However, the core identification challenge, often called the fundamental problem of causal inference, is that for any given unit, only one of $Y_i(1)$ and $Y_i(0)$ can be observed, never both. Note that trying two different assignments on the same unit sequentially may not work either, since the nature of the unit may have changed in the meantime.

The core insight of Neyman (Neyman, 1923) was that although Δ_i itself is unobservable, randomization allows us to learn averages of these effects. In finite samples, define the sample average treatment effect (SATE):

$$\Delta = \frac{1}{n}\sum_{i=1}^{n}(Y_i(1) - Y_i(0)).$$

Difference-in-Means Estimation.

A natural estimator for measuring the causal impact of a treatment is the treated-control **d**ifference in sample means:

$$\hat{\tau}_{DM} = \frac{1}{n_1}\sum_{W_i=1} Y_i - \frac{1}{n_0}\sum_{W_i=0} Y_i, \qquad n_w = |\{i : W_i = w\}|.$$

To study this estimator, we impose two further assumptions. First, SUTVA (the Stable Unit Treatment Value Assumption):

$$Y_i = Y_i(W_i), \qquad i = 1, \ldots, n.$$

SUTVA implies two key conditions. First, it means there is no interference between units, so that unit i's outcome depends only on W_i, not on others' assignments. Second, it implies that there are no hidden versions of the treatment, meaning the intervention is uniform enough that $Y_i(1)$ represents a single, unambiguous potential outcome for each unit, rather than varying effects from different doses or versions of the treatment. Together, these rule out spillovers and allow us to cleanly interpret the observed treated-control contrast as a causal contrast between $Y_i(1)$ and $Y_i(0)$.

The second assumption is that the treatment is completely randomized conditional on a fixed number of treated units n_1:

$$\mathbb{P}(W_i = 1 \mid \{Y_i(0), Y_i(1)\}_{i=1}^{n}, n_1) = \frac{n_1}{n}, \qquad i = 1, \ldots, n.$$

These assumptions yield the following foundational result:

Theorem 4.1.1 (Difference in means is unbiased in a completely randomized trial)**.** *Under the assumptions above,*

$$\mathbb{E}[\hat{\tau}_{DM} \mid \{Y_i(0), Y_i(1)\}_{i=1}^{n},\ n_0 > 0,\ n_1 > 0] = \Delta.$$

The proof is short and relies on linearity of expectation, so we give it here:

Proof. When $n_1 > 0$, we have that

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{n_1}\sum_{W_i=1} Y_i\right] &= \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^{n} W_i Y_i \;\middle|\; \{Y_i(0), Y_i(1)\}_{i=1}^{n},\ n_1\right] \\
&= \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^{n} W_i Y_i(1) \;\middle|\; \{Y_i(0), Y_i(1)\}_{i=1}^{n},\ n_1\right] \quad \text{(SUTVA)} \\
&= \frac{1}{n_1}\sum_{i=1}^{n} Y_i(1)\,\mathbb{E}[W_i \mid \{Y_i(0), Y_i(1)\}_{i=1}^{n},\ n_1] \quad (Y_i(1) \text{ is fixed in the conditioning set}) \\
&= \frac{1}{n}\sum_{i=1}^{n} Y_i(1). \quad \text{(Randomization)}
\end{aligned}$$

An analogous argument gives

$$\mathbb{E}\left[\frac{1}{n_0}\sum_{W_i=0} Y_i\right] = \frac{1}{n}\sum_{i=1}^{n} Y_i(0),$$

and subtracting the two equations above yields the desired result. □

4.2 Recommended Reading

Historically, the potential-outcomes framework for causal inference traces its origins to Jerzy Neyman's 1923 work on agricultural experiments (See the 1990 English translation of the original, (Neyman, 1923)). Although Neyman and R.A. Fisher were developing foundational experimental methodologies during the same era (with Fisher formulating his randomization-based design methodology (Fisher, 1935)), their early efforts were independent. Both were separately motivated by the practical demands of early 20th-century agricultural research stations: Neyman at the Institute of Agriculture in Bydgoszcz, Poland (Encyclopaedia Britannica, 2026a), and Fisher at the Rothamsted Experimental Station in the UK (Box, 1978). Decades later, Donald Rubin modernized and extended Neyman's framework to accommodate both randomized and nonrandomized observational settings (Rubin, 1974); see also the excellent review in (Holland, 1986). Rubin is also explicitly credited with formalizing the Stable Unit Treatment Value Assumption (SUTVA) (Rubin, 1980). Theorem 4.1.1 and the framing leading up to it is based on Chapter 1 of (Wager, 2024).

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